

Linear Programming

Q1.

1. Max : $0.095A + 0.08B + 0.09C + 0.09D + 0.09E$

Subject to : $A + B + C + D + E = \$100,000$

$B + E \geq \$50,000$

$A + D + E \leq \$50,000$

$A + B + E \geq \$30,000$

$0.095A + 0.08B + 0.09D \geq 0.4(0.095A + 0.08B + 0.09C + 0.09D + 0.09E)$

$A, B, C, D, E \geq 0$

2.

3. $A = \$20,339$ $B = \$20,339$ $C = \$29,661$ $D = 0$ $E = \$29,661$

Max return = $\$8,898$

Q2.

1. Max : $(21 - 17.25)X_{1R} + (21 - 15.75)X_{2R} + (21 - 17.75)X_{3R} + (25 - 17.25)X_{1S}$
 $+ (25 - 15.75)X_{2S} + (25 - 17.75)X_{3S}$

X_{iR} = number of barrels to be use to produce regular gasoline

X_{iS} = number of barrels used to produces supreme gasoline

Subject to :

1) $X_{1R} + X_{2R} + X_{3R} = 300$

2) $X_{1S} + X_{2S} + X_{3S} = 450$

3) $X_{1R} + X_{1S} \leq 150$

4) $X_{2R} + X_{2S} \leq 350$

5) $X_{3R} + X_{3S} \leq 300$

6) $100X_{1R} + 87X_{2R} + 110X_{3R} / 300 \geq 90$

7) $100X_{1S} + 87X_{2S} + 110X_{3S} / 450 \geq 97$

8) $X_{ij} \geq 0$

2.

3. $X_{1R} = 0$, $X_{2R} = 260.87$, $X_{3R} = 39.13$. $X_{1S} = 80.77$, $X_{2S} = 119.23$, $X_{3S} = 250$

Maximum profit is $\$50,125.00$

Q3.

1. X_i = number of boards to which cutting pattern i is applied.

$$\text{Min: } X_1 + X_2 + X_3 + X_4 + X_5 + X_6$$

Subject to:

$$1) 3X_1 + 2X_2 + 2X_3 + X_4 \geq 5000$$

$$2) X_2 + 2X_4 + X_5 \geq 1200$$

$$3) X_3 + X_5 + 2X_6 \geq 300$$

$$3. X_1 = 1267 \quad X_3 = 300 \quad X_4 = 600 \quad \text{Total} = 2167$$

4. New function would be:

$$\text{Min: } (25-21)X_1 + (25-23)X_2 + (25-25)X_3 + (25-25)X_4 + (25-20)X_5 + (25-22)X_6$$

Q4.

1. Excel sensitivity report gives an allowable decrease of 4 for X_3 - this is the amount the variable coefficient can fall by without changing the optimal configuration. The smallest value X_3 can take is therefore 0, as $X_3 \geq 0$

2. ?

3. Feasible region would decrease because this is a minimisation problem, and increasing the RHS of the first constraint increases the minimum value the variables can take. Objective function is increased at a rate of 2 per unit increase in the first constraint's RHS. $4 + 2 \times 5 = 14$.

$$4. 4 - 2 \times 1 = 2.$$

Q5.

1. Min : $260X_{13} + 220X_{14} + 290X_{15} + 230X_{23} + 240X_{24} + 310X_{25}$

where X_1 = Eustis, X_2 = Clermont, X_3 = Miami, X_4 = Orlando, X_5 = Tallahassee

Subject to :

1) $X_{13} + X_{14} + X_{15} \leq 20$

2) $X_{23} + X_{24} + X_{25} \leq 20$

3) $X_{13} + X_{23} \geq 10$

4) $X_{14} + X_{24} \geq 15$

5) $X_{15} + X_{25} \geq 10$

6) $X_{ij} \geq 0$

2.

3. $X_{14} = 10$ $X_{15} = 10$ $X_{23} = 10$ $X_{24} = 5$ Total cost = \$8600

6. No change as the optimal solution uses 15 tons of Clermont produce, thus is unaffected by capacity falling to 16 tons

7. Reducing the capacity in Eustis forces us to substitute Clermont produce for these 5 units, at an additional cost of \$20 per unit. Functions new value is \$8700

8. Unsure on this question.

Q6.

1. Generators provide more profit than alternators, and require less inputs, so Maximising generator production will Maximise profits.

Generator : $2X_1 + X_2$ where X_1 = wiring and X_2 = testing

Max : $2X_1 + X_2$ st. $2X_1 \leq 260$, $X_1 \leq 130$, $X_2 \leq 140$

130 generators requires $260X_1$ and $130X_2$

No excess wiring capacity, 10 excess testing capacity

2. $2X_1 \leq 270$, $X_1 \leq 135$, $X_2 \leq 140$

5 extra generators

~~\$2~~ $135 \times \$250 = \$33,750$ profit